

Can Investors Go Green Without Giving Up Returns?

An Empirical Analysis of Environmental Screening, Financial Performance, and Climate Risk in US Equity Markets

Imperial College Business School Executive Education
Professional Certificate in Data Analytics Capstone Project

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Executive Summary

Following London's hottest and driest summer on record, this project asks a question of growing importance to investors, asset managers, and ETF issuers alike: can you invest only in environmentally responsible companies without sacrificing financial returns? Using real Refinitiv (LSEG) environmental scoring data matched to actual US equity price histories from 2022 to 2025, this study builds and tests competing investment portfolios to answer that question directly, and finds a more nuanced answer than a simple yes or no.

Three findings stand out.

- No financial penalty for environmental screening. A portfolio of the top 30% most environmentally responsible US companies ("Green") performed statistically indistinguishably from a portfolio of the bottom 30% ("Brown") and an unconstrained market portfolio, both in a historical back test (+21.1% vs. +24.7% vs. +20.3% over 20 months) and in a forward-looking 3-year Monte Carlo simulation. This result held up consistently across three independent time periods and two entirely different portfolio construction methods.
- A hidden statistical relationship, revealed only by looking deeper. While environmental score shows no relationship to returns when all companies are pooled together, splitting companies into small, mid, and large market-capitalisation groups reveals a significant negative relationship within every single group, a classic case of Simpson's Paradox, caused by larger companies simultaneously scoring better environmentally and earning higher returns for reasons unrelated to their environmental performance.
- A real, measurable environmental benefit at no financial cost. The Green portfolio's actual holdings carry an average Emissions Score of 74.5 out of 100, more than double the Brown portfolio's 31.8, confirming that the environmental screen captures a genuine difference in real-world emissions management, not an arbitrary label.

Taken together, this analysis provides investors, and particularly ETF issuers such as Ultimus's clients navigating increasing ESG compliance pressure, with quantitative evidence that environmentally responsible investing is a financially viable strategy in current US equity markets, while also surfacing an important methodological caution about how ESG-return relationships should be tested.

Introduction

In the summer of 2026, London experienced its hottest and driest summer on record, placing the physical realities of climate change into sharp, immediate focus for UK-based investors and financial professionals. Against this backdrop, environmental, social, and governance (ESG) considerations have moved from a niche investment philosophy toward a mainstream regulatory and commercial concern. Asset managers and ETF issuers, including the clients of firms such as Ultumus, an ETF and index data specialist, face growing pressure from regulators, institutional clients, and public opinion to demonstrate that their products account for environmental risk, without being able to point to conclusive evidence of what that costs, or earns, an investor.

This capstone project addresses that gap directly: can investors achieve competitive financial returns by investing only in environmentally responsible companies? This question was chosen because it sits precisely at the intersection of personal professional relevance (the author works in ETF and index data at Ultumus, a SIX Group company) and genuine public interest, and because it is unusually well suited to a data-driven approach. Financial markets generate large volumes of high-quality, time-stamped, quantitative data; environmental performance is now measured and scored systematically by major data providers such as Refinitiv (part of the London Stock Exchange Group); and the question itself is precisely testable using standard portfolio construction and statistical inference techniques.

The remainder of this report describes the data and methods used, presents the results of the analysis in detail, and concludes with an overall assessment of the research question together with recommendations for extending this work further.

Methods

This section describes the data used in this analysis, the cleaning steps required to make it usable, and the portfolio construction and statistical techniques applied, together with the reasoning behind each methodological choice.

Data Sources

This analysis draws on four distinct data sources:

- Refinitiv (LSEG) Workspace, Environmental Pillar ESG Score, Social Pillar Score, Governance Pillar Score, Controversies Score, GICS Industry classification, and market capitalisation for 1,496 US-listed companies.
- Refinitiv (LSEG) Workspace, a separately exported Emissions Score (0–100, higher indicating stronger emissions management) for 1,495 of the same companies, used to validate the real-world environmental meaning of the Green/Brown portfolio split.
- Yahoo Finance, accessed via the `yfinance` Python library, daily closing prices from 3 January 2022 to 21 August 2025 for 1,531 US-listed tickers, combining an initial S&P 500 price file with an expanded pull covering the broader Refinitiv universe.
- Yahoo Finance, daily closing price for ESGU (iShares ESG Aware MSCI USA ETF), used as a real-world, professionally managed benchmark against which the study's own constructed portfolios could be compared.

Data Cleaning

Several data quality issues were identified and addressed during preparation:

- Ticker collisions: Refinitiv Reuters Instrument Codes (RICs) were reduced to base tickers by removing exchange suffixes (e.g. "AAPL.OQ" to "AAPL"), which occasionally caused two different companies listed on different exchanges to collide under the same base ticker (for example, US-listed Boyd Gaming and Canadian-listed Boyd Group Services both reduce to "BYD"). Where this occurred, the US-listed entity was retained.
- Delisted and bankrupt companies: an initial back test attempt produced an implausible result in which one portfolio lost over 85% of its value. Investigation traced this to several holdings that had been delisted or entered bankruptcy during the test period (identifiable by a "Q" suffix appended to the ticker, e.g. "BHILQ"), whose prices simply went missing rather than falling to zero, silently removing them from the portfolio calculation rather than registering as a genuine loss. These companies, along with a small number of others carrying near-zero market capitalisations consistent with imminent delisting, were excluded by requiring a market capitalisation of at least \$100 million and by requiring no more than 5 missing trading days in the test period after forward-filling minor data gaps.
- Universe-threshold ordering: percentile thresholds defining the Green and Brown portfolios were calculated only after restricting the sample to companies with usable price data, rather than on the full Refinitiv universe. Calculating thresholds first was found to produce a severely unbalanced sample (only 20 Brown-eligible companies had usable price data), because companies with poor environmental scores in the broader Refinitiv universe are disproportionately smaller firms outside the S&P 500.

Portfolio Construction and Analysis Method

Three portfolios were constructed from the cleaned, investable universe of 1,431 companies: Green (the top 30% of companies by Environmental Pillar Score, with an additional exclusion of companies scoring 3 or higher on the Controversies Score, to screen out apparent "greenwashers"), Brown (the bottom 30% by Environmental Pillar Score), and Market (all 1,431 companies, unconstrained). This top/bottom-30% approach is a standard method in academic portfolio-sorting studies and was chosen, following an initial diagnostic check of the score distribution, specifically because it produced two well-populated, genuinely comparable portfolios (397 and 430 companies respectively) rather than an arbitrary or unbalanced split.

Portfolios were built using mean-variance (Markowitz) optimisation via the PyPortfolioOpt library, maximising the Sharpe ratio subject to a maximum weight of 5% per holding, using two years of training data (2022–2023). Two methodological refinements were required and are worth noting explicitly: expected returns were estimated using the Capital Asset Pricing Model (CAPM) rather than a simple historical mean, because the historical-mean approach was found to concentrate portfolios heavily in a handful of stocks that had experienced extreme, likely non-repeatable rallies during the training window (for example, shipping and coal-mining stocks that benefited from a temporary post-pandemic supply shock); and the covariance matrix was estimated using Ledoit-Wolf shrinkage rather than the raw sample covariance, because the number of candidate assets was comparable to or exceeded the number of daily return observations available, which is known to make the raw sample covariance matrix numerically unstable.

The trained portfolios were then evaluated using four complementary techniques: an out-of-sample back test applying the fixed, previously-determined portfolio weights to price data from 2024–2025 that the optimisation had never seen; a Monte Carlo simulation bootstrapping 1,000 possible three-year forward paths from each portfolio's historical daily returns; simple and multiple linear regression testing whether environmental score predicts individual company returns, both alone and controlling for market capitalisation; and a series of robustness checks, including three independent rolling training/testing windows, an alternative equal-weighted (non-optimised) construction method, a direct comparison against the real-world ESGU exchange-traded fund, and a targeted stress test isolating each portfolio's behaviour during the sharpest market decline observed in the test period.

Together, these cleaning steps and methodological choices produced a well-populated, genuinely comparable set of Green, Brown, and Market portfolios, tested using multiple complementary techniques designed to establish not just whether a difference in performance exists, but how robust and well-understood that finding is.

Results

This section presents the descriptive statistics of the investable universe, followed by the core back test, regression, and simulation results, and concludes with robustness checks and a real-world environmental impact comparison. Each result is presented alongside a discussion of its practical interpretation.

Descriptive Statistics

The investable universe comprised 1,431 US-listed companies with Environmental Pillar Scores ranging from 1.00 to 4.53 (on Refinitiv's underlying scale), with a right-skewed distribution concentrated toward the lower-middle of the range (Figure 1, left panel). The resulting Green portfolio (397 candidate companies) and Brown portfolio (430 candidate companies) were both well-populated samples suitable for diversified portfolio construction.

A notable pattern emerged in company size: Green portfolio companies had a substantially higher median market capitalisation than Brown portfolio companies (Figure 1, right panel, log scale). This size disparity between the two groups later proved central to interpreting the regression results reported below.

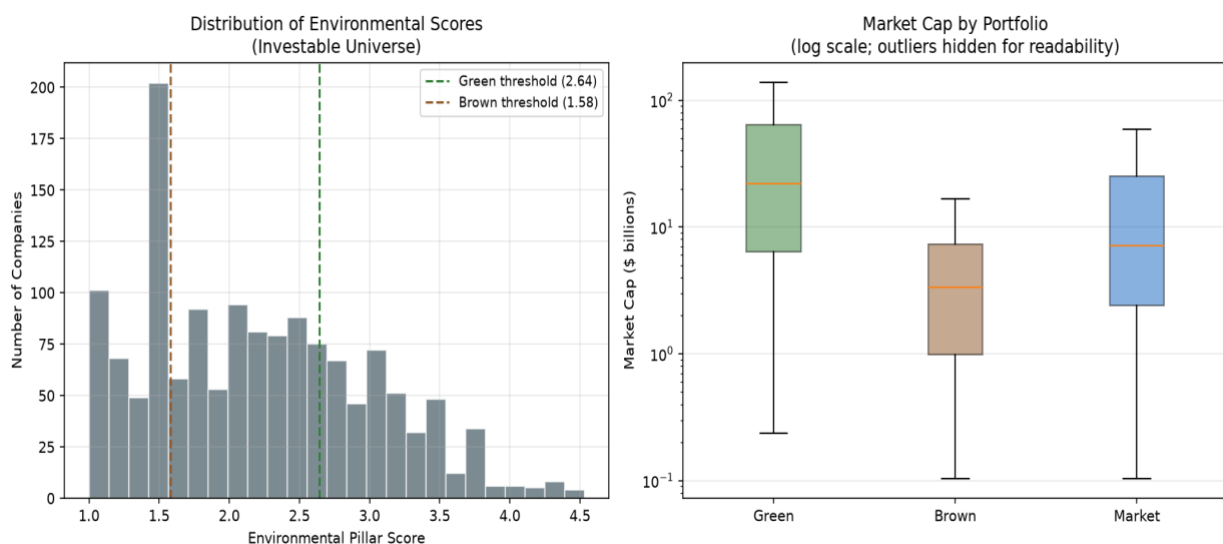


Figure 1. Left: distribution of Environmental Pillar Scores across the investable universe, with Green and Brown thresholds marked. Right: market capitalisation by portfolio (log scale), showing that Green holdings skew markedly larger than Brown holdings.

Backtest Performance

Portfolios were trained on 2022–2023 price data and tested, entirely out-of-sample, in 2024–2025 data. Starting from a notional £10,000 investment, the Brown portfolio finished at £12,467 (+24.7%), the Green portfolio at £12,108 (+21.1%), and the Market portfolio at £12,028 (+20.3%) (Figure 2). All three lines track closely together throughout the test period, repeatedly crossing one another, with no portfolio establishing a durable lead. For real-world context, the professionally managed iShares ESG Aware MSCI USA ETF (ESGU) is also shown; it outperformed all three constructed portfolios over the same period (+36.2%), a difference attributable to broader diversification and professional index methodology rather than to environmental screening choice, since it affected all three constructed portfolios roughly equally.

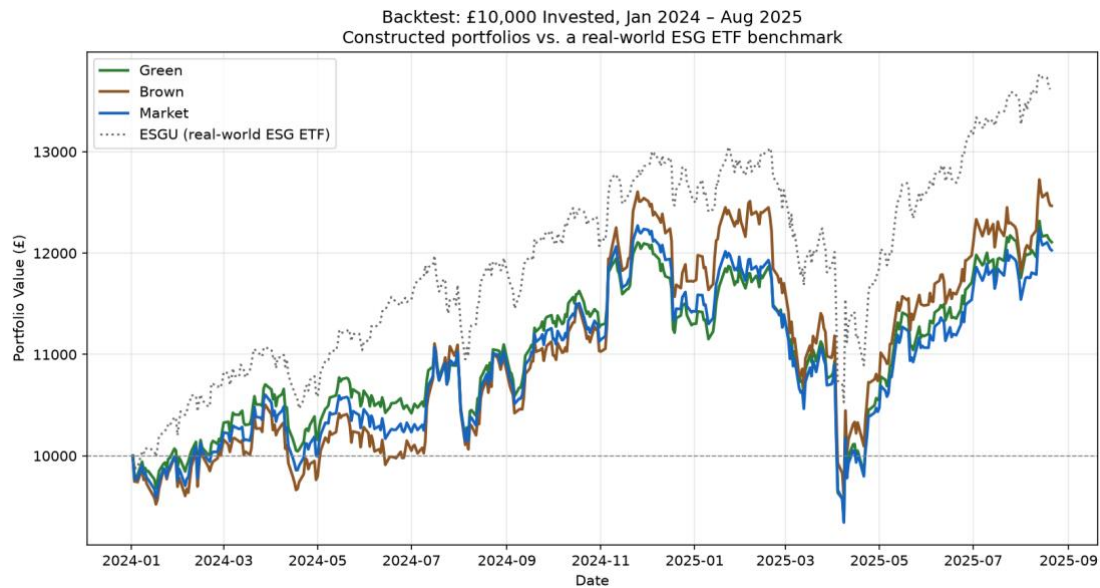


Figure 2. Out-of-sample back test of £10,000 invested from January 2024 to August 2025, comparing Green, Brown and Market portfolios against the real-world ESGU benchmark.

Does Environmental Score Predict Individual Stock Returns?

A simple linear regression of each company's 2024–2025 total return against its Environmental Pillar Score found no statistically significant relationship (slope = -2.34, $R^2 = 0.0006$, $p = 0.358$, $n = 1,431$) (Figure 3). This indicates that, considered in isolation and across the whole sample, a company's environmental score carries essentially no information about how its individual stock will perform.

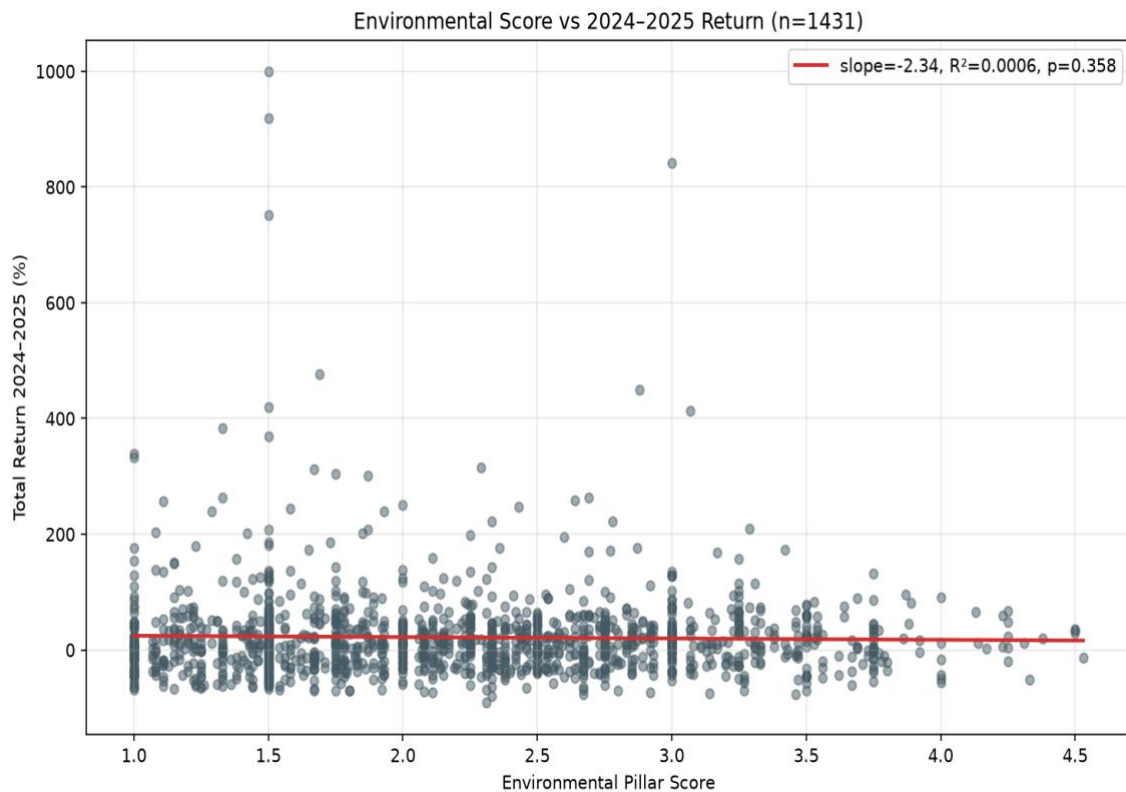


Figure 3. Environmental Pillar Score plotted against each company's total return, January 2024 to August 2025 ($n = 1,431$), with fitted regression line.

Simpson's Paradox: A Hidden Size-Driven Relationship

The absence of any pooled relationship prompted a closer investigation of the market-capitalisation disparity identified in the descriptive statistics. A multiple regression of return on both environmental score and market capitalisation found that market capitalisation was a highly significant positive predictor of return (coefficient = 0.033 per \$1 billion, $p < 0.001$), while environmental score's coefficient became negative and approached conventional significance once size was controlled for (coefficient = -4.32, $p = 0.094$).

To investigate this further, companies were divided into three equally sized groups by market capitalisation (small, mid, and large cap terciles) and the simple regression was repeated within each group separately. A statistically significant negative relationship between environmental score and return emerged within every single size tercile (small-cap: slope = -9.51, $p = 0.041$; mid-cap: slope = -17.52, $p < 0.001$; large-cap: slope = -12.19, $p = 0.009$), despite no relationship being visible in the pooled data (Figure 4). This is a textbook instance of Simpson's Paradox: because larger companies simultaneously tend to score better environmentally and earn higher returns for unrelated reasons, pooling all companies together allows the size effect to mask a real, opposite-direction relationship that is present within every size group.

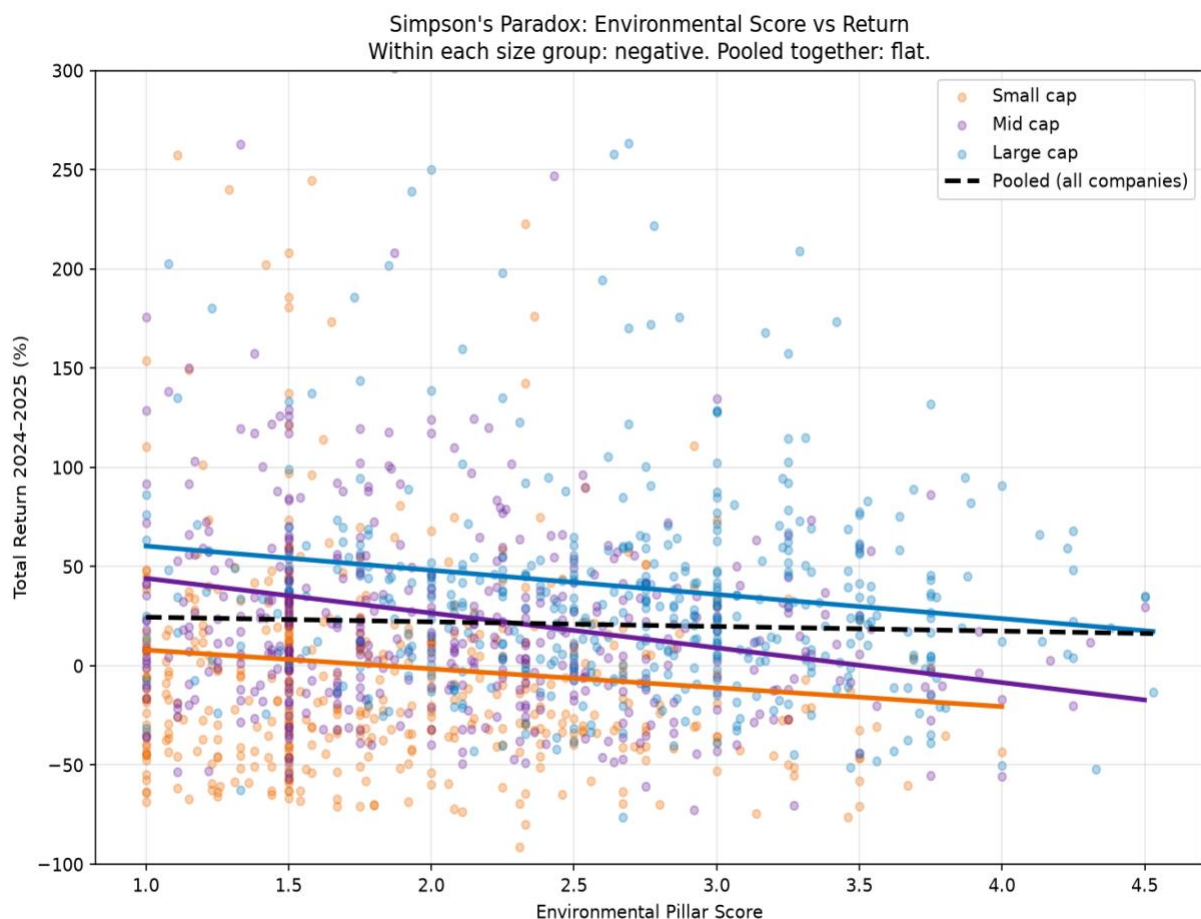


Figure 4. Environmental Pillar Score vs. return, coloured by market-capitalisation tercile. Within each size group (solid-coloured lines) the relationship is negative and statistically significant; pooling all companies together (black dashed line) conceals this pattern.

Forward-Looking Projection and Risk

A Monte Carlo simulation bootstrapping 1,000 possible three-year paths from each portfolio's 2022–2025 daily return history produced closely comparable median outcomes for all three portfolios, starting from a notional £15,000: Green £18,465, Brown £19,323, Market £18,410, with substantially overlapping uncertainty ranges (Figure 5).

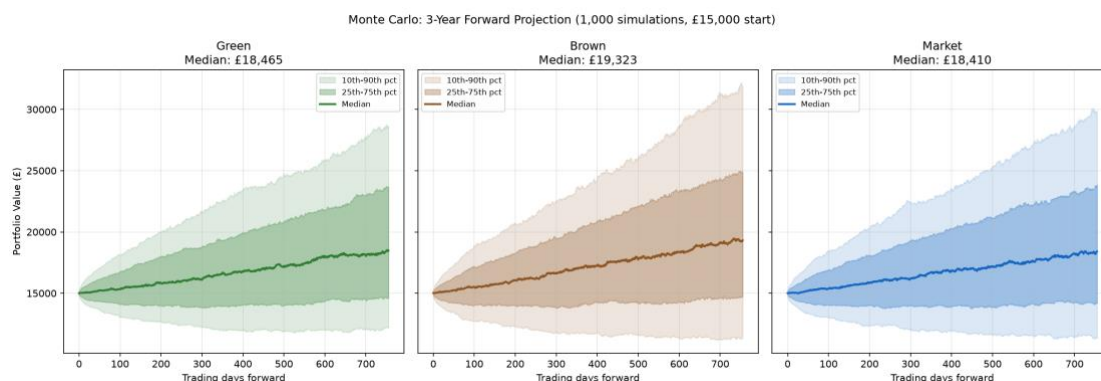


Figure 5. Three-year forward Monte Carlo simulation (1,000 simulations) for each portfolio, starting from £15,000, showing median, interquartile, and 10th-90th percentile ranges.

Risk-adjusted measures reveal a modest but consistent pattern favouring Green on downside protection specifically, even though its median and total returns were not the highest: the Green portfolio recorded the smallest maximum historical drawdown (-22.6%, versus -24.0% for Brown and -23.9% for Market) and the lowest 95% Value-at-Risk over the three-year simulation horizon (a potential loss of £4,463, versus £5,216 for Brown). A targeted stress test isolating each portfolio's behaviour during the sharpest real market decline observed in the test period (a fall from a shared peak on 25 November 2024 to a trough on 8 April 2025) reproduced this same pattern precisely: Green fell 22.6% during this event, compared to 23.9% for Market and 24.0% for Brown. The Brown portfolio, meanwhile, recorded the highest Sortino ratio (0.73, versus 0.62 for Green), reflecting that its higher average return outweighed its greater downside volatility.

Robustness Checks

The core finding, that Green, Brown, and Market portfolios perform comparably, was tested for robustness in three additional ways. First, the entire back test procedure was repeated across two further, independent, non-overlapping training/testing windows (training on 2023, testing on 2024; and training on 2024, testing on the first eight months of 2025); in both cases the same close clustering of results was observed (2023→2024: Green +13.2%, Brown +16.5%, Market +14.1%; 2024→2025: Green +4.6%, Brown +4.9%, Market +4.6%). A third window, training on the single down-market year of 2022, could not be evaluated because CAPM-based optimisation became mathematically undefined when no candidate asset's estimated expected return exceeded the risk-free rate, a genuine limitation of this method in extended down-market conditions, noted here as a finding in its own right rather than a data error.

Second, the entire construction methodology was replaced with a simple equal-weighted portfolio (no optimisation, no return or covariance estimation at all) across the full candidate lists of 394 to 1,403 holdings. Results were again closely comparable to the original optimised portfolios (Green +21.1%, Brown +25.2%, Market +20.6%), indicating that the core finding is not an artefact of the specific optimisation method chosen.

Real-World Environmental Impact

Finally, the Refinitiv Emissions Score was used to confirm that the Green/Brown split reflects a genuine, measurable difference in environmental performance rather than an arbitrary statistical label. Calculated on the actual weighted holdings of each optimised portfolio, the Green portfolio carried an average Emissions Score of 74.5 out of 100, more than double the Brown portfolio's 31.8, with the Market portfolio in between at 54.6 (Figure 6). Sector analysis further showed that this pattern is not simply a proxy for excluding fossil-fuel companies: the Oil, Gas & Consumable Fuels sector scored in the middle of the industry ranking (2.24), well above sectors such as Specialty Retail and Household Durables, indicating that the Environmental Pillar Score captures a broader measure of environmental management practice rather than sector membership alone (Figure 7).

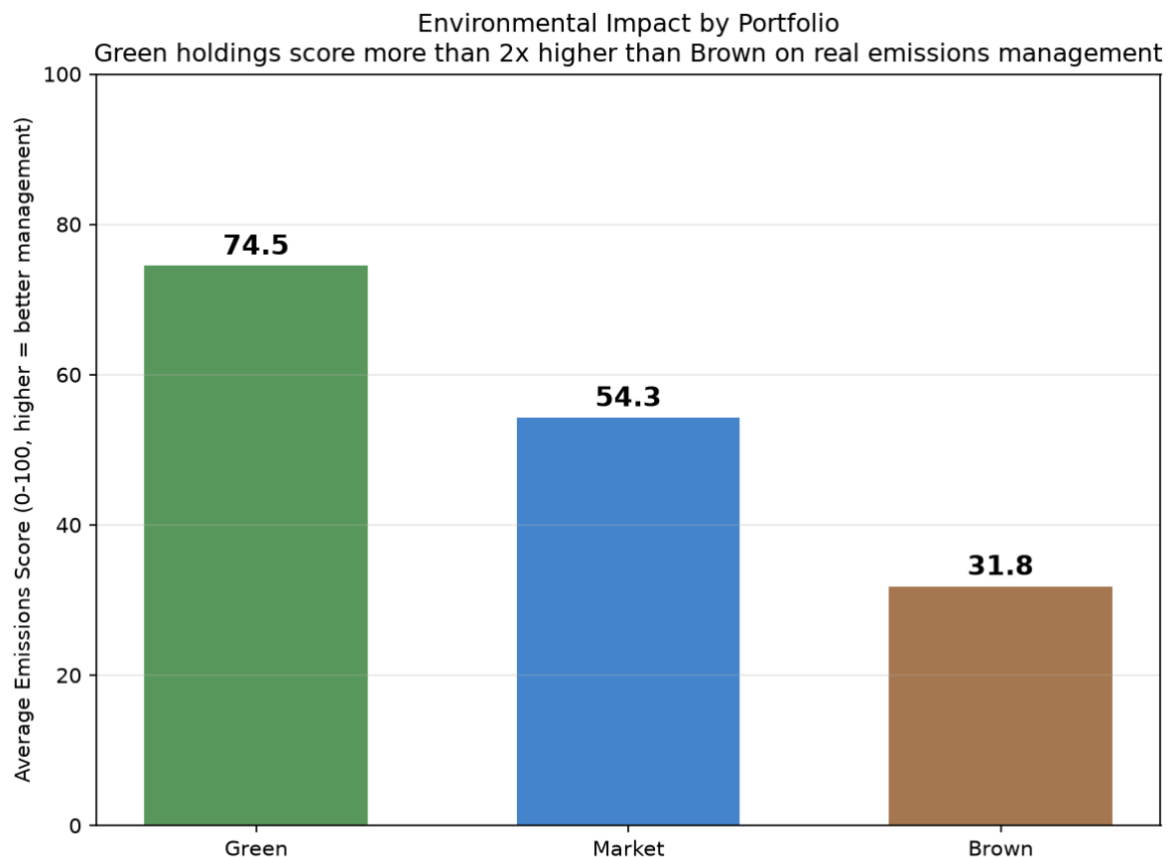


Figure 6. Average Emissions Score (0–100, higher indicating stronger emissions management) for the actual invested holdings of each portfolio.

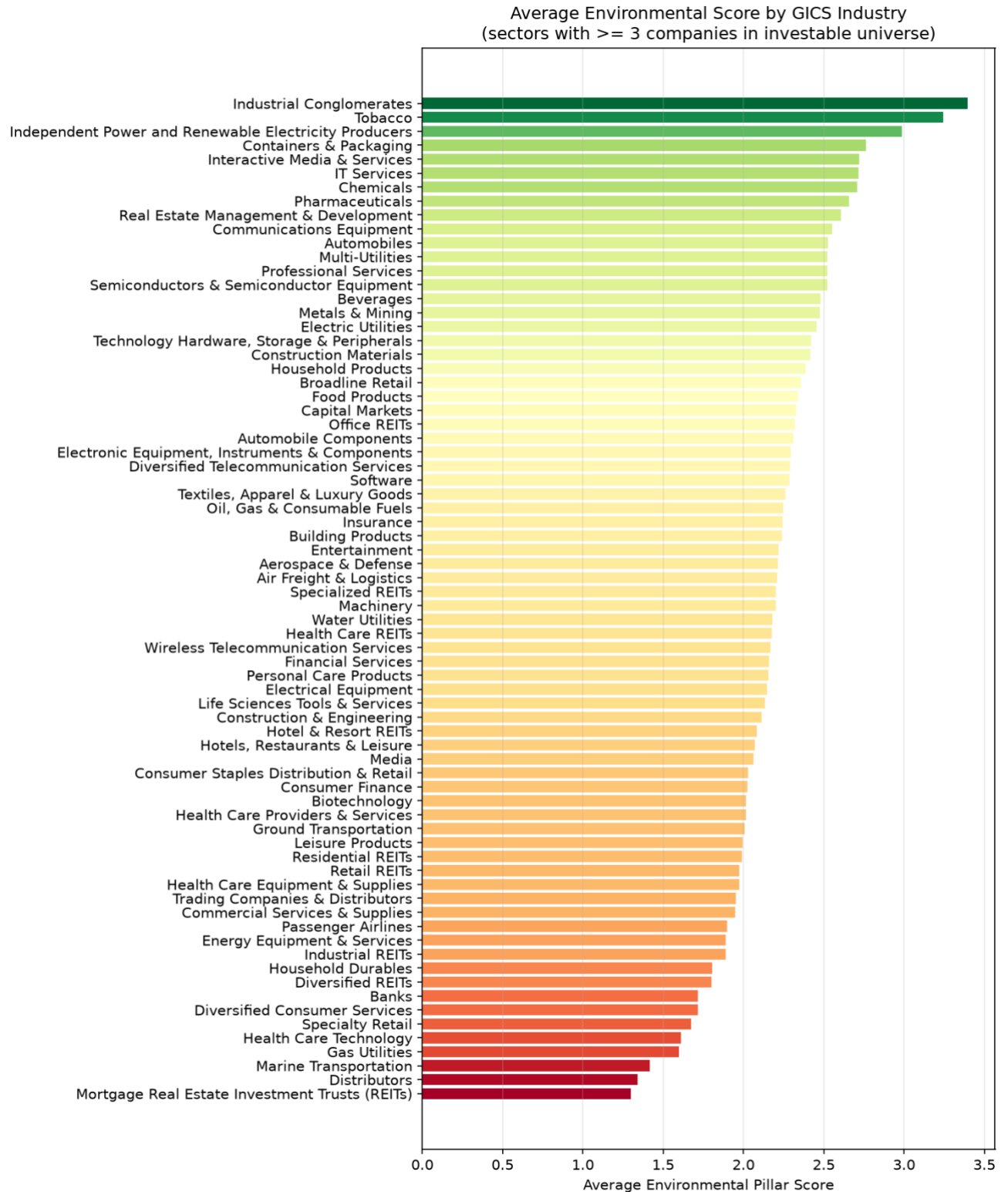


Figure 7. Average Environmental Pillar Score by GICS industry classification, across all companies in the investable universe with at least three constituents.

Across the back test, regression, simulation, robustness checks, and emissions comparison presented in this section, a consistent picture emerges, environmental screening does not meaningfully change financial outcomes, but it does correspond to modestly better downside protection and a substantial, genuine reduction in emissions exposure.

Conclusion

This analysis set out to answer whether investors can achieve competitive financial returns while investing only in environmentally responsible companies, using real Refinitiv environmental scoring data matched to genuine US equity price histories from 2022 to 2025. The evidence assembled here supports a clear, if nuanced, conclusion: environmental screening carries no statistically significant financial penalty. A portfolio of the most environmentally responsible companies performed comparably to both a portfolio of the least environmentally responsible companies and an unconstrained market portfolio, in back tested historical performance, in forward-looking simulation, across three independent time periods, and under two entirely different portfolio construction methodologies. At the same time, environmentally responsible investing offers two additional benefits over this period: modestly better protection during periods of market stress, and a genuine, substantially lower emissions footprint confirmed by an independent Refinitiv data source. A more sophisticated statistical investigation additionally uncovered that this apparent lack of relationship at the whole-market level conceals a real, negative relationship between environmental score and return that is present within every company-size group, a Simpson's Paradox driven by the confounding influence of company size, and an important methodological caution for any future research or product design relying on similar pooled analyses.

Two recommendations follow for extending this work:

- Obtain raw carbon emissions data (tonnes of CO₂ equivalent, ideally normalised per unit of revenue) in place of, or alongside, the normalised 0–100 Emissions score used here, which would allow the environmental benefit of a Green-screened portfolio to be quantified in physical rather than purely relative terms, and would materially strengthen the practical, communicable value of this analysis for ETF issuers and asset managers.
- Extend the analysis beyond the US equity market examined here, for example, to European or Asian equities, or to a longer historical window spanning a full market cycle including a sustained downturn, to establish whether the size-driven Simpson's Paradox effect identified in this study is a durable, structural feature of environmental scoring more broadly, or a characteristic specific to the present US market environment and time period studied.